

HANDS-ON 1

Task	Hint
1. Open a terminal and identify if you are a normal user or a superuser Try to switch to root (superuser) and report what happens	<code>whoami</code> <code>su</code>
2. Identify your shell	<code>echo \$SHELL</code>
3. Check the system distribution Check the kernel version Check the glibc version	<code>uname -a</code> <code>uname -r</code> <code>ldd --version</code>

HANDS-ON 2

Task	Hint
1. Open a terminal and list your files and directories.	<code>ls</code>
2. Check more options about the <code>ls</code> command and display all files.	<code>ls --help</code> <code>ls -a</code>
3. Clear the screen.	<code>Clear</code>
4. Compare the information given by different kinds of help about <code>pwd</code> command.	<code>whatis pwd</code> <code>help pwd</code> <code>man pwd</code> <code>info pwd</code>
5. Review a few used commands with arrow and compare with output from history.	<code>history</code>
6. Print the <code>date</code> to the screen. Save date into <code>date.txt</code> file. Add another line with date into that file. Display more information about <code>date.txt</code> file.	<code>date</code> <code>date > date.txt</code> <code>date >> date.txt</code> <code>stat date.txt</code>
7. Exit the shell.	<code>Exit</code> or <code>ctrl+D</code>
8. Start terminal again. Print the current directory. Navigate to the <code>/bin</code> directory. Print the current directory. Go to the <code>/usr</code> directory. List the content of the current directory. Go back to previous directory. List all files starting with “col” and sort them alphabetically.	<code>pwd</code> <code>cd /bin</code> <code>pwd</code> <code>cd /usr</code> <code>ls</code> <code>cd -</code> <code>ls col* sort</code>

Go back to your home directory using the **absolute path**.

List all the directories starting from the root ("/") directory . Try to guess what kind of files can be stored there.

How much disk space does the /bin folder take?

How much storage space is available on the file system?

```
cd /home/<username>
```

```
ls /
```

(see Linux tree in slides)

```
du -cksh /bin
```

```
df -h /
```

HANDS-ON 3

Task	Hint
1. Open a terminal. Make a directory called "LinuxIntro" in your home directory. Copy the directory /usr/share/zoneinfo/Europe/ to the created LinuxIntro directory. Make a directory called "test" in the LinuxIntro directory. Make a directory called "test1" in the LinuxIntro directory. Rename test1 to test2. Copy all files from /usr/share/zoneinfo/Europe/ to the created LinuxIntro/test directory. Change the filename of Brussels at ~/LinuxIntro/test/ to "Chocolate_o_Clock" in directory. Try to print the contents of ~/Chocolate_o_Clock. Report what happens. Get the file type of Chocolate_o_Clock. Copy the file Chocolate_o_Clock from ~/LinuxIntro/test to your home directory. Create a symbolic link called "chocolate_link" to the Chocolate_o_Clock file in your home directory.	<pre>mkdir ~/LinuxIntro cp -r /usr/share/zoneinfo/Europe/ ~/LinuxIntro mkdir ~/LinuxIntro/test mkdir ~/LinuxIntro/test1 mv ~/LinuxIntro/test1 ~/LinuxIntro/test2 cp -r /usr/share/zoneinfo/Europe/* ~/LinuxIntro/test mv ~/LinuxIntro/test/Brussels ~/LinuxIntro/test/Chocolate_o_Clock cat ~/LinuxIntro/test/Chocolate_o_Clock file ~/LinuxIntro/test/Chocolate_o_Clock cp ~/LinuxIntro/test/Chocolate_o_Clock ~/Chocolate_o_Clock ln -s ~/Chocolate_o_Clock chocolate_link</pre>

Get the filetype of the linked file. Remove <code>~/Chocolate_o_Clock</code>. Try to print the contents of <code>chocolate_link</code>. Try again to get the filetype of the <code>chocolate_link</code>. Remove the broken symlink.	<pre>file chocolate_link</pre> <pre>rm -i ~/Chocolate_o_Clock</pre> <pre>cat ~/chocolate_link</pre> <pre>file ~/chocolate_link</pre> <pre>rm ~/chocolate_link</pre>
2. Clear the screen. Go to the <code>LinuxIntro</code> directory. Download the file https://raw.githubusercontent.com/hpcleuven/Linux-intro/master/tabel.dat and https://raw.githubusercontent.com/hpcleuven/Linux-intro/master/matstats.log. Show the content of the file <code>tabel.dat</code> and <code>matstats.log</code>. Show the last part and the first part of <code>matstat.log</code> (show 30 lines) Check if the patterns "ardaa" and "ardbb" appear in <code>matstats.log</code>. List the files of the directory in <code>less</code>. Check if the file <code>tabel.dat</code> is in the list.	<pre>clear</pre> <pre>cd ~/LinuxIntro</pre> <pre>wget https://raw.githubusercontent.com/hpcleuven/Linux-intro/master/tabel.dat</pre> <pre>wget https://raw.githubusercontent.com/hpcleuven/Linux-intro/master/matstats.log</pre> <pre>cat tabel.dat (or less tabel.dat or more tabel.dat)</pre> <pre>less matstats.log (or more matstats.log)</pre> <pre>tail -30 matstats.log</pre> <pre>head matstats.log</pre> <pre>grep ardaa matstats.log</pre> <pre>grep ardbb matstats.log</pre> <pre>ls -al less</pre> <pre>ls -al grep tabel.dat</pre>

3. Create a file called "test4edit". Open the file with either <code>nano</code> or <code>vi</code>. Write something in the file. Close the file and print the contents.	<pre>touch test4edit nano test4edit vi test4edit cat test4edit</pre>
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HANDS-ON 4

Task	Hint
1. Open a terminal. Go to the <code>LinuxIntro</code> directory. Check if the <code>Europe</code> directory is still there. Archive the <code>Europe</code> directory. Archive and gzip the <code>Europe</code> directory. Create a new directory "newtest" under <code>LinuxIntro</code> and unpack the archive into it.	<pre>cd ~/LinuxIntro ls Eur* tar -cf Euro.tar Europe/ tar -czf ico.tar.gz Europe/ mkdir newtest cd newtest cp ../Europe.tar.gz . tar -xzf Europe.tar.gz</pre>
2. Go back to the <code>LinuxIntro</code> directory. Create a directory "testfiles", create in it a file "file1" containing a few numbers. Copy <code>file1</code> to "file2". Remove the user write permission to <code>file2</code>. Print the files and permissions in <code>testfiles</code> directory. Try to edit <code>file2</code> and add some numbers. Does it work?	<pre>cd ~/LinuxIntro mkdir testfiles cd testfiles echo "1 2 3" > file1 cp file1 file2 chmod u-w file2 ls -la nano file2</pre>

remove write access for the user of the `testfiles` directory and print the files and permissions in the current directory.

Try to copy `file1` in that directory to a file `"file3"`. What happens?

Remove all access to `testfiles` for the others and print the files and permissions in the current directory.

Try to list the files in `testfiles` directory.

Remove read access for the user for `testfiles` directory and print the files and permissions in the current directory.

Try to list the files in `testfiles` directory. What happens?

Go to the `testfiles` directory and go back one level up.

Remove execute access for the user for `testfiles` directory, print the files and permissions in the current directory.

Try to go to the `testfiles` directory. What happens?

```
cd ..  
chmod u-w testfiles  
ls -la
```

```
cp testfiles/file1  
testfiles/file3
```

```
chmod o-rwx testfiles  
ls -la
```

```
ls testfiles
```

```
chmod u-r testfiles  
ls -la
```

```
ls testfiles
```

```
cd testfiles  
cd ..
```

```
chmod u-x testfiles  
ls -la
```

```
cd testfiles
```